

Functional Outcome Following Blood Pressure Drops Within 24 Hours After Successful Mechanical Thrombectomy

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Abstract

Background and Objectives

The optimal target blood pressure (BP) during and after mechanical thrombectomy (MT) for acute ischemic stroke remains uncertain. While BP drops during MT are associated with worse outcomes, evidence on postprocedural BP instability is limited. We aimed to assess the impact of BP drops within the first 24 hours after MT on functional outcome.

Methods

Retrospective observational study analyzing data from 2 tertiary stroke centers, gathered between October 2020 and September 2022 within the German Stroke Registry Endovascular Treatment, a national prospective registry of patients receiving MT for ischemic stroke. High-resolution BP data of patients who reached successful recanalization at the end of MT (modified Thrombolysis in Cerebral Infarction [mTICI] $\geq 2b$) were obtained using stroke unit monitoring. BP drops were defined as systolic BP decreases of ≥ 40 mm Hg occurring within 1 hour during the first 24 hours after groin puncture. Primary outcome was level of disability at 3 months, assessed on the modified Rankin Scale (mRS shift analysis). Secondary outcomes included early neurologic deterioration (defined as an increase of 4 or more points on the National Institutes of Health Stroke Scale (NIHSS) between admission and 24-hour assessment) and death at 3 months. Associations between BP drops and functional outcomes were evaluated using multivariable ordinal and binary logistic regression.

Results

Among 300 patients analyzed (median age 77 years [interquartile range (IQR) 65–83], 46.0% female, median admission NIHSS 13 points [IQR 7–17]), 123 patients (41.0%) had at least 1 drop, including 77 patients (62.6%) with 1 drop and 46 patients (37.4%) with 2 or more drops. BP drops were independently associated with worse functional outcome, both in binary (drops vs no drops; adjusted common odds ratio [OR] 1.66 [95% CI 1.02–2.70]) and count-based analysis (adjusted common OR per +1 drop; 1.33 [95% CI 1.05–1.69]). Patients with drops more often had early neurologic deterioration (22.2% vs 13.4%; adjusted OR 2.22 [95% CI 1.03–4.78]). BP drops were not associated with death at 3 months (adjusted OR 1.25, 95% CI 0.66–2.40).

Discussion

BP drops within 24 hours after successful MT are associated with worse functional outcome. Targeted measures to prevent or mitigate BP drops should be explored in future studies.

Introduction

In acute ischemic stroke, blood pressure (BP) is frequently elevated, with most patients presenting with systolic values above 140 mm Hg at admission.¹ This acute hypertensive response

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Glossary

acOR = adjusted common OR; **aOR** = adjusted OR; **ASPECTS** = Alberta Stroke Program Early CT Score; **BP** = blood pressure; **GSR-ET** = German Stroke Registry Endovascular Treatment; **ICA** = internal carotid artery; **ICH** = intracranial hemorrhage; **IQR** = interquartile range; **mRS** = modified Rankin Scale; **MT** = mechanical thrombectomy; **mTICI** = modified Thrombolysis in Cerebral Infarction; **NIHSS** = NIH Stroke Scale; **OR** = odds ratio; **RCT** = randomized controlled trial; **sICH** = symptomatic ICH.

is usually transient and may reflect both preexisting hypertension and a physiologic reaction to cerebral ischemia.^{2,3} Although elevated BP increases the risk of hemorrhagic transformation, it also helps sustain perfusion of the ischemic penumbra.^{4,5} This dual role raises a central clinical dilemma: Does BP reduction early after acute ischemic stroke improve outcomes, or does it increase the risk of exacerbating ischemia by interfering with autoregulatory mechanisms?³

In patients undergoing mechanical thrombectomy (MT) for ischemic stroke, international guidelines recommend keeping BP below 180/105 mm Hg during and 24 hours after MT.^{6,8} However, this recommendation has been adopted in an arbitrary manner from BP management of randomized controlled trials in the context of IV thrombolysis (IVT).^{6,7} Observational research suggests that very high BP (>220/120 mm Hg) is associated with worse outcomes and higher rates of symptomatic intracranial hemorrhage after MT.⁹⁻¹² On the other hand, 2 randomized controlled trials (RCT) have shown that intensive BP lowering to targets below 140 mm Hg (Outcome in Patients Treated with Intraarterial Thrombectomy - OptiMAL Blood Pressure Control [OPTIMAL-BP])¹³ or even below 120 mm Hg (Second Enhanced Control of Hypertension and Thrombectomy Stroke Study [ENCHANTED-2])¹⁴ led to increased disability at 3 months. The Blood Pressure Target in Acute Stroke to Reduce Hemorrhage After Endovascular Therapy trial failed to demonstrate a benefit of intensive BP lowering (<130 mm Hg) with regard to the occurrence of parenchymal hemorrhage,¹⁵ and the Blood Pressure After Endovascular Stroke Therapy-II trial suggested a low probability of benefit from low systolic BP targets if tested in future large trials.¹⁶ While aggressive BP lowering seems to be harmful, the overall quality of evidence on the optimal BP target range in patients with MT is low, and consequently, strength of recommendation is weak in both European and American guidelines.^{6,7}

In addition to maximum and minimum BP thresholds, the trajectory of BP over time may be of importance. Observational data suggest that BP drops during MT may worsen outcome, and therefore, BP stability may be of relevance.¹⁷⁻²¹ Indeed, BP stability was treatment target in the Individualized Blood Pressure Management During Endovascular Stroke Treatment (INDIVIDUATE) trial,²² but INDIVIDUATE failed to demonstrate a benefit in actively stabilizing BP during MT (admission BP $\pm$ 10 mm Hg). The Effect of Individualized Versus Standard Blood Pressure Management During

Mechanical Thrombectomy for Anterior Ischemic Stroke (DETERMINE) trial,²³ which also compared individually stable vs standard BP management during MT, similarly yielded neutral results (presented at European Stroke Organization Conference [ESOC] 2025, not yet published). Although RCT evidence is limited, the European Stroke Organization (ESO) / European Society of Minimally Invasive Neurological Therapy (ESMINT) guidelines recommend avoiding excessive BP drops in patients undergoing MT.⁸ While these recommendations focus on BP management during MT, evidence for optimal BP control after the procedure remains limited. We aimed to evaluate whether the guideline recommendation holds true for the postprocedural period and sought to analyze the impact of BP drops within the first 24 hours after MT on functional outcome.

Methods

Study Design and Study Population

We conducted a retrospective observational study analyzing data from 2 tertiary stroke centers, collected within the German Stroke Registry - Endovascular Treatment (GSR-ET), a national, multicenter, prospective observational registry described in detail before.^{24,25} All consecutive individuals 18 years or older who received MT for ischemic stroke in its participating centers are included in the registry.²⁴ Functional status at 3 months is regularly assessed through telephone interviews using the modified Rankin Scale (mRS).²⁶

For this analysis, all patients who received MT between October 2020 and September 2022 at the 2 centers were screened for eligibility. BP values were measured as part of the automated stroke unit monitoring, digitized through the COPRA documentation system (version 6.2, COPRA System GmbH, Berlin, Germany), and transferred to the electronic medical record. These data were subsequently extracted in bulk from the Charité Data Warehouse²⁷ and checked for completeness. In case of missing BP values, patient records were reassessed manually, and the respective BP values were gathered, if available. The period of interest was the time from groin puncture to 24 hours thereafter.

In line with the randomized controlled trials on BP management after MT, patients had to achieve modified Thrombolysis in Cerebral Infarction (mTICI) score of 2b or 3 at the end of MT to be included.^{13,14} Moreover, continuous BP

monitoring had to be available for at least 50% of the time window of interest.

BP management was performed as per local standards of care, which included keeping systolic BP <180 mm Hg after MT. Urapidil, dihydralazin, dexmedetomidine, and clonidine (in this sequence) were used in case of threshold exceedance. IV vasoconstrictors were used on a case-by-case decision, mostly in patients with BP-dependent clinical deterioration or hemodynamic instability.

Variables and Outcomes

The ESO/ESMINT guidelines on BP management in patients undergoing MT recommend the avoidance of “excessive” BP drops,⁸ yet they leave “excessive” undefined. Given the absence of an established definition in the literature, we pragmatically defined a BP drop as a ≥ 40 mm Hg decrease in systolic BP within 1 hour, a magnitude we considered likely to be regarded as “excessive” by most clinicians. BP was recorded with high temporal resolution, from the first documented measurement after groin puncture up to 24 hours postprocedure, either intermittently with a BP cuff or continuously through arterial line. Implausible BP measurements (systolic BP <50 or >300 mm Hg; diastolic BP <20 or >150 mm Hg) were excluded. In patients with noninvasive BP monitoring, BP was measured every 30 or every 60 minutes. If multiple readings were available within a 30-minute window, values were averaged to minimize bias from single failed or invalid measurements. In case of monitoring gaps exceeding 1 hour (e.g., during MRI), drop assessment resumed with the next 2 valid recordings. In patients with invasive BP monitoring, values were averaged at 30-minute intervals, and consecutive averaged values were compared. BP drops were defined as a decline of ≥ 40 mm Hg between 2 consecutive BP measurements, using the immediately preceding BP value as the reference.

Abrupt BP fluctuations in patients with arterial line were analyzed by converting the raw data to 1-minute intervals and calculating the following metrics as BP changes between the median of 2 consecutive minutes (mm Hg/minute, respectively): (1) overall time rate, including both BP increases and decreases over time, (2) average drop rate, depicting the velocity of drops ≥ 40 mm Hg toward their lowest point, and (3) average decline rate, including any downward movement >10 mm Hg.

We collected the following variables: admission systolic and diastolic BP, admission heart rate, time from groin puncture to first documented BP value, time from groin puncture to first BP drop, mean, maximum, and minimum systolic and diastolic BP during the 24-hour period. Furthermore, we assessed the absolute value of the first BP drop (in mm Hg) and its proportion relative to the directly preceding BP value (in percent), the number of BP drops, and number of BP peaks (defined as an increase in systolic BP of ≥ 40 mm Hg within 1 hour). Data on concurrent antihypertensive medication administered during the 24-hour period were collected.

The National Institutes of Health Stroke Scale (NIHSS) was used to assess stroke severity on admission and at 24 hours. Alberta Stroke Program Early CT Score (ASPECTS) was applied to quantify early ischemic changes on admission imaging. Functional status at 3 months was evaluated using the mRS, reaching from 0 (no residual stroke symptoms) to 6 (death). Procedure adverse events such as vasospasms (any procedure-related vasoconstriction, as documented by the neurointerventionalist), periprocedural clot migration, and dissection/perforation were recorded.

The primary outcome was functional status at 3 months (mRS shift analysis). As secondary outcomes, we assessed functional independence at 3 months (mRS ≤ 2), dependency with unassisted ambulation at 3 months (mRS ≤ 3), early neurologic improvement (decrease of at least 4 points on the NIHSS or NIHSS score of 0 at 24 hours), and early neurologic deterioration (increase on the NIHSS of at least 4 points between admission and 24 hour assessment). We further included any intracranial hemorrhage (ICH) at 24-hour follow-up imaging, symptomatic ICH (sICH) as defined in the European Cooperative Acute Stroke Study-II (blood at any site of the brain with an NIHSS increase of at least 4 points), in-hospital death, and death at 3 months as secondary outcomes.

Statistical Analysis

Continuous baseline variables, treatment times, and BP values are presented as mean ($\pm$ SD) or median and interquartile range [IQR], depending on variable distribution. Dichotomous variables are reported as absolute numbers and percentages. For descriptive analyses, we compared (1) patients with at least 1 BP drop to patients with no BP drop and (2) within the subgroup of patients with at least 1 BP drop, patients with 1 drop vs patients with multiple drops. The groups were compared using the χ^2 test for dichotomous variables and Student *t* test or Mann-Whitney *U* test for continuous variables, as appropriate.

The primary outcome, disability at 3 months (mRS), was analyzed using ordinal regression, with BP drops assessed either binary (any vs none) or as a count variable (per additional drop) and odds ratios (ORs) >1 indicating a worse functional outcome. Adjustments were made for variables selected a priori based on their clinical relevance and status as well-known major confounders, namely, age, sex, mRS before the index event, NIHSS at baseline, time from last-seen-well (or symptom onset) to hospital admission, IVT, and ASPECTS (including posterior circulation; PC-ASPECTS). For the secondary and safety outcomes, binary logistic regression analyses were conducted, adjusting for the abovementioned factors. All analyses were performed using IBM SPSS Statistics for Windows (version 30.0.0.0, IBM Corp., Armonk, NY).

Sensitivity and Supplementary Analyses

Reperfusion status at the end of MT (mTICI) as a modifying factor of BP drops was assessed by restricting analyses to

patients with complete reperfusion at the end of MT (mTICI3). Furthermore, to assess the impact of BP variability independent of discrete drop events, we conducted multivariable regression analyses using SD of systolic BP values as variable of interest. The impact of different BP drop definitions (20, 25, 30, 35 mm Hg) was assessed performing multivariable ordinal regression for the primary outcome, in both binary and count-based analysis.

Standard Protocol Approvals, Registrations, and Patient Consents

The GSR-ET registry was centrally approved by the Ethics Committee of the Ludwig-Maximilians University LMU, Munich (689-15).²⁴ Informed consent was waived due to the retrospective nature of the study.

Reporting Guidelines

The reporting of this observational study is in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.

Data Availability

Anonymized data not published within this article will be made available by request from any qualified investigator.

Results

Baseline Demographics

From 397 patients who underwent MT between October 2020 and September 2022 at the 2 tertiary hospitals, continuous BP data were available in 364 individuals (92%). Successful recanalization (mTICI 2b/3) was reached in 300 patients (82.4%), which constituted the final study population (Figure 1). The median age was 77 years (IQR 65–83); the median NIHSS 13 points (IQR 7–17) and 46.0% were female. Invasive BP monitoring was used in 14.0%, while 86.0%

received measurement through BP cuff. BP drops occurred in 123 patients (41.0%). Of those, 77 patients (62.6%) had 1 drop, and 46 patients (37.4%) had 2 or more drops.

Patients with BP drops were 1 year older (median age 77 [IQR 69–85] years vs 76 [IQR 63–82] years, $p = 0.03$); sex and stroke severity did not differ between groups (Table 1). An internal carotid artery (ICA)-stenosis >70% was twice as common in patients with BP drops (22.8% vs 10.7%, $p < 0.01$). Prestroke disability (mRS before the index event), cardiovascular risk factors, stroke etiology, and ASPECTS on admission imaging were equally distributed (Table 1). Rates of IVT, time from last-seen-well to hospital arrival, distribution of vessel occlusion site, and time from hospital arrival to groin puncture (door to groin time) were all balanced between groups. General anesthesia was used in 97.5% (drops) and 93.6% (no-drops), respectively ($p = 0.12$). Complete recanalization (mTICI 3) was reached in 68.3% (drops) and 71.2% (no-drops; $p = 0.59$). Rates of procedural complications such as dissection/perforation, clot migration, and vasospasms did not differ (see Table 1 for further details). When comparing patients with 2 or more drops to patients with 1 drop, baseline characteristics, admission imaging, and treatment details did not differ (eTable 1).

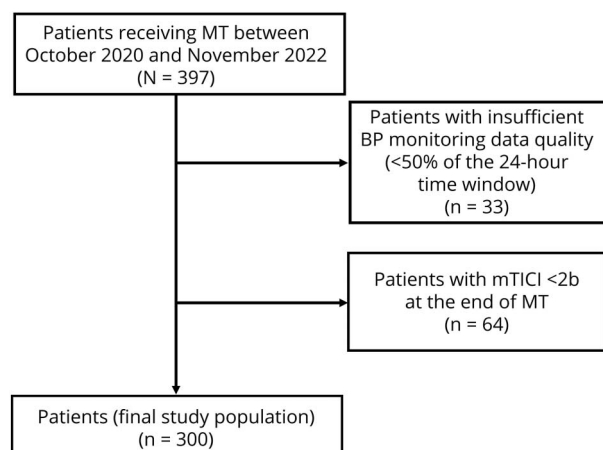
BP Details

At hospital admission, systolic BP was higher in patients with BP drops (152 ± 26 vs 145 ± 21 mm Hg, $p = 0.02$). Diastolic BP and heart rate did not differ (see Table 2 for details). Invasive BP monitoring was used in 11.4% of patients with drops and 15.8% of patients without drops ($p = 0.28$). The median BP-monitoring coverage of the 24-hour period was 96% (IQR 92–100) in patients with drops and 92% (IQR 85–100) in patients without drops ($p < 0.001$). The median time from groin puncture to first available BP measurement was 71 minutes (IQR 0–123) in patients with drops and 94 minutes (IQR 44–146) in patients without drops ($p = 0.01$). The median absolute drop amplitude was 47 mm Hg (IQR 43–54); the median relative drop was 30.0% of the preceding BP value (IQR 27.0%–33.9%). In patients with invasive BP assessment, the mean (SD) overall time rate of BP changes (both increases and decreases) was 4.3 ± 1.14 mm Hg/min in patients with drops and 3.20 ± 1.21 mm Hg/min in patients without drops ($p < 0.01$). The mean (SD) average drop rate was 1.8 ± 0.9 mm Hg/min. For further details on BP dynamics, see eTable 2. High-resolution patient-level data on BP coverage and time to occurrence of drops are presented in eFigure 1 in the Supplement. Details on mean, minimum, and maximum BP values of patients with and without drops as well as the occurrence of BP peaks are presented in Table 2. Detailed BP data stratified by number of drops (1 drop vs 2 or more drops) are presented in eTable 3.

Concurrent Medication

Data on antihypertensive drugs were available in 236 patients (78.7% of the study sample). Oral antihypertensives were given in 38.6% of patients, without any difference between groups (drops vs no drops). IV vasodilators were

Figure 1 Patient Population



MT = mechanical thrombectomy; mTICI = modified Thrombolysis in Cerebral Infarction

Table 1 Baseline Demographics and Treatment Variables

	All (n = 300)	No BP drops (n = 177)	BP drops (n = 123)	p Value
Age, y, median (IQR)	77 (65–83)	76 (63–82)	77 (69–85)	0.03
Female sex, n (%)	138 (46.0)	77 (43.5)	61 (49.6)	0.30
Pre-mRS score ≤2, n (%)	246 (82.0)	148 (83.6)	98 (79.7)	0.38
NIHSS at admission, median (IQR)	13 (7–17)	12 (7–18)	14 (8–17)	0.15
Hypertension, n (%)	92 (74.8)	135 (76.3)	92 (74.8)	0.77
Diabetes mellitus, n (%)	69 (23.0)	41 (23.2)	28 (22.8)	0.94
Dyslipidemia, n (%)	109 (36.3)	67 (37.9)	42 (34.1)	0.51
Atrial fibrillation, n (%)	156 (52.0)	88 (49.7)	68 (55.3)	0.34
Current smoking, n (%)	58 (19.3)	39 (22.0)	19 (15.4)	0.16
Anticoagulation, n (%)	71 (23.7)	44 (24.9)	27 (22.0)	0.56
Antiplatelet agent, n (%)	87 (29.0)	56 (31.6)	31 (25.2)	0.23
Stroke etiology, n (%)				
Cardioembolic	165 (55.0)	98 (55.4)	67 (54.5)	0.88
Large-artery atherosclerosis	110 (36.7)	64 (36.2)	46 (37.4)	0.83
Dissection	3 (1.0)	3 (1.7)	0	0.15
Other determined	5 (1.7)	4 (2.3)	1 (0.8)	0.34
Undetermined	16 (5.3)	8 (4.5)	8 (6.5)	0.45
Admission imaging, n (%)				
CT	277 (92.3)	165 (93.2)	112 (91.1)	0.49
MRT	23 (7.7)	12 (6.8)	11 (8.9)	
ASPECTS, median (IQR)	8 (7–10)	8 (7–10)	9 (7–10)	0.37
Perfusion-based imaging, n (%)	103 (34.3)	60 (33.9)	42 (35.0)	0.85
ICA stenosis >70%, n (%)	47 (15.7)	19 (10.7)	28 (22.8)	<0.01
Occluded vessel, n (%)				
Nonterminal ICA (ICA-I)	30 (10.0)	16 (9.0)	14 (11.4)	0.38
Terminal ICA (ICA-T)	34 (11.3)	20 (11.3)	14 (11.4)	
Proximal MCA M1	61 (20.3)	42 (23.7)	19 (15.4)	
Distal MCA M1	74 (24.7)	45 (25.4)	29 (23.6)	
MCA M2	59 (19.7)	36 (20.3)	23 (18.7)	
ACA	3 (1.0)	1 (0.6)	2 (1.6)	
VA	6 (2.0)	3 (1.7)	3 (2.4)	
BA	26 (8.7)	10 (5.6)	16 (13.0)	
PCA	7 (2.3)	4 (2.3)	3 (2.4)	
Occlusion territory, n (%)				0.11
Left anterior	49 (39.8)	80 (45.2)	49 (39.8)	
Right anterior	52 (42.3)	86 (48.6)	55 (44.7)	
Posterior	39 (13.0)	11 (6.2)	19 (15.4)	
Known onset of stroke, n (%)	149 (49.7)	95 (53.1)	57 (46.3)	0.25

Continued

Table 1 Baseline Demographics and Treatment Variables (*continued*)

	All (n = 300)	No BP drops (n = 177)	BP drops (n = 123)	p Value
IV thrombolysis, n (%)	109 (36.3)	62 (35.0)	47 (38.2)	0.57
Primary admission at interventional hospital, n (%)	263 (87.7)	154 (87.0)	109 (88.6)	0.68
Last-seen-well or symptom onset to admission, min, median (IQR)	155 (72–418)	146 (69–418)	174 (75–414)	0.35
Admission to thrombolysis, min, median (IQR)	36 (25–55)	37 (26–58)	36 (23–55)	0.52
General anesthesia, n (%)	277 (95.2)	160 (93.6)	117 (97.5)	0.12
Admission to groin puncture, min, median (IQR)	90 (65–135)	92 (68–138)	86 (62–133)	0.24
Groin puncture to flow restoration, min, median (IQR)	30 (19–52)	30 (17–49)	30 (20–57)	0.25
Complete reperfusion (mTICI 3), n (%)	210 (70.0)	126 (71.2)	84 (68.3)	0.59
Dissection/perforation, n (%)	8 (2.7)	4 (2.3)	4 (3.3)	0.60
Clot migration, n (%)	29 (9.7)	16 (9.0)	13 (10.6)	0.66
Vasospasms, n (%)	37 (12.3)	21 (11.9)	16 (13.0)	0.77
ICA stenting, n (%)	26 (8.7)	11/19 (57.9)	12/28 (42.9)	0.31

Abbreviations: ACA = anterior cerebral artery; ASPECTS = Alberta Stroke Program Early CT Score; BA = basilar artery; BP = blood pressure; ICA = internal carotid artery; IQR = interquartile range; MCA = middle cerebral artery; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale; PCA = posterior cerebral artery; VA = vertebral artery.

administered more often in patients with BP drops (51.1% vs 34.9%, $p = 0.01$). For further details, including active substances, see eTable 4.

Clinical and Safety Outcomes

BP drops were strongly associated with worse clinical outcome at 3 months (mRS shift) both analyzed binary (any vs none; adjusted common OR [acOR] 1.66 [95% CI 1.02–2.70]; $p = 0.04$) and as count (acOR per additional drop: 1.33 [95% CI 1.05–1.69]; $p = 0.02$). Both the odds of functional independence (mRS ≤ 2 ; 28.2% vs 50.3% adjusted OR [aOR] 0.35 [95% CI 0.18–0.69]; $p < 0.01$) and dependency with unassisted ambulation (mRS ≤ 3 ; 38.5% vs 59.3%; aOR 0.41 [95% CI 0.22–0.78]; $p < 0.01$) at 3 months were lower in patients with drops (vs no drops). In addition, BP drops were associated with early neurologic deterioration (22.2% vs 13.4%; aOR 2.22 [95% CI 1.03–4.78]; $p = 0.04$). Although there was no overall association between BP drops and the occurrence of any ICH, higher odds of any ICH were observed when taking the number of drops into account (no drops 20.3%, ≥ 2 drops 30.4%; aOR per additional drop 1.31 [95% CI 1.01–1.71]; $p = 0.04$). The odds of sICH were similar between groups (5.1% vs 6.4%; aOR 0.94 [95% CI 0.29–3.09]; $p = 0.92$). Although in-hospital mortality (21.1% vs 15.3%) and mortality at 3 months (33.3% vs 25.1%) were numerically higher in patients with BP drops, no independent association was found (in-hospital: aOR 1.66 [95% CI 0.79–3.49]; $p = 0.18$; at 3 months: aOR 1.25 [95% CI 0.66–2.40]; $p = 0.50$). Details on all outcomes and analyses according to the number of BP drops are presented in Table 3 and eTable 5. The distribution of mRS scores by BP drop status (any vs none) is shown in Figure 2. Figure 3 shows outcomes according to the number of BP drops.

Sensitivity Analyses

When restricting analyses to patients with complete reperfusion (mTICI 3, $n = 210$), we found that the association between BP drops and worse clinical recovery was numerically even more pronounced than in the whole study population (mRS shift; drops vs no drops; acOR 1.86 [95% CI 1.04–3.33]; $p = 0.04$). Any ICH, on the other hand, was not associated with BP drops in these patients. For details on further outcomes, see eTable 6.

Increased BP variability, measured as SD of systolic BP values, was independently associated with worse clinical recovery (mRS shift: acOR per +1 mm Hg 1.09 [1.04–1.14]; $p < 0.001$). Associations of SD with secondary outcomes were consistent with the results on BP drops (eTable 7). When defining BP drops at different thresholds (20, 25, 30, and 35 mm Hg), we noted that there was no longer an association between BP drops and primary outcome in binary analysis (any drops vs none). In count-based analyses, however, even drops as low as 20 mm Hg were associated with worse functional recovery, if they occurred repeatedly. With lower definition of BP drops, the point estimate per +1 drop gradually decreased (acOR for 20 mm Hg per +1 drop: 1.19 [95% CI 1.06–1.33]; $p < 0.01$). For details, see eTable 8.

Discussion

In this retrospective observational study leveraging high-resolution stroke unit monitoring data, we report the following key findings: First, BP drops ≥ 40 mm Hg during the first 24 hours after MT are common, with about 4 in 10

Table 2 Blood Pressure Details

	All (n = 300)	No drops (n = 177)	Any drops (n = 123)	p Value
Systolic BP at admission, mm Hg, mean $\pm$ SD	148 $\pm$ 23	145 $\pm$ 21	152 $\pm$ 26	0.02
Diastolic BP at admission, mean $\pm$ SD	85 $\pm$ 17	83 $\pm$ 17	87 $\pm$ 18	0.11
HR at admission, bpm, mean $\pm$ SD	83 $\pm$ 21	83 $\pm$ 20	82 $\pm$ 22	0.61
Invasive BP monitoring, n (%)	42 (14.0)	28 (15.8)	14 (11.4)	0.28
Time from groin puncture to first documented BP value, min, median (IQR)	86 (1–140)	94 (44–146)	71 (0–123)	0.01
Blood pressure monitoring coverage for 24 h, %, median (IQR)	92 (88–100)	92 (85–100)	96 (92–100)	0.01
Mean systolic BP, mm Hg, mean $\pm$ SD	132 $\pm$ 14	130 $\pm$ 14	135 $\pm$ 14	<0.01
Mean diastolic BP, mm Hg, mean $\pm$ SD	69 $\pm$ 12	69 $\pm$ 12	69 $\pm$ 13	0.92
Maximum systolic BP, mm Hg, mean $\pm$ SD	174 $\pm$ 28	166 $\pm$ 28	185 $\pm$ 25	<0.001
Maximum diastolic BP, mm Hg, mean $\pm$ SD	107 $\pm$ 21	105 $\pm$ 21	111 $\pm$ 21	0.03
Minimum systolic BP, mm Hg, mean $\pm$ SD	93 $\pm$ 18	97 $\pm$ 17	89 $\pm$ 18	<0.001
Minimum diastolic BP, mm Hg, mean $\pm$ SD	46 $\pm$ 13	48 $\pm$ 13	44 $\pm$ 13	0.02
Median (IQR) drop amplitude, mm Hg			47 (43–54)	—
Median (IQR) drop amplitude, %			30.2 (27.0–33.9)	—
Time from groin puncture to first drop, min, median (IQR)	—	—	390 (210–780)	—
First drop within 360 min after groin puncture, n (%)	—	—	61 (49.6)	—
No. of drops, n (%)				
1			77 (62.6)	
2			25 (20.3)	
≥ 3			21 (17.0)	
Any peaks, n (%)	115 (38.3)	27 (15.3)	88 (71.5)	<0.001
No. of peaks, n (%)				
1	66 (22.0)	22 (12.4)	44 (35.8)	<0.001
2	26 (8.7)	5 (2.8)	21 (17.1)	
3	15 (5.0)	—	15 (12.2)	
>3	8 (2.7)	—	8 (6.5)	

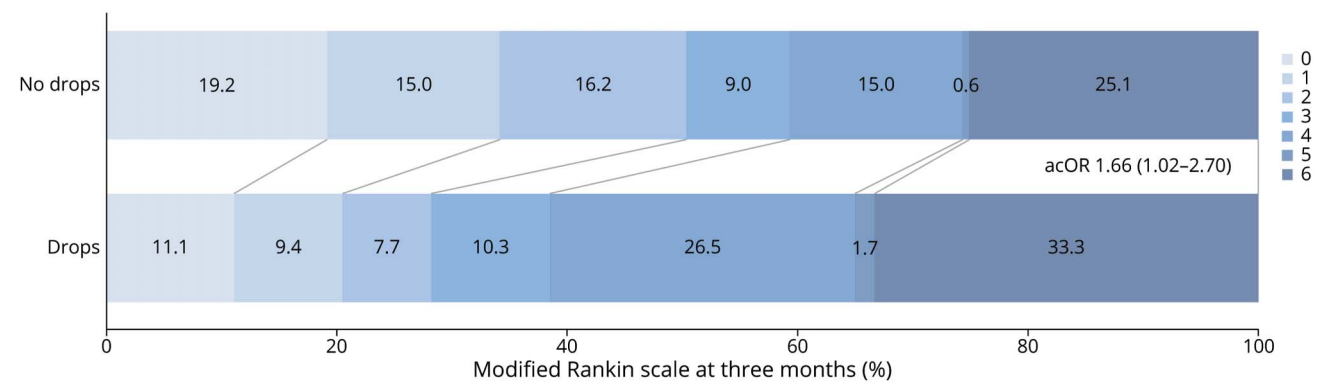
Abbreviations: BP = blood pressure; HR = heart rate; IQR = interquartile range.

patients experiencing at least 1 BP drop. Second, BP drops during stroke unit monitoring were associated with early neurologic deterioration. Third and most importantly, BP drops were associated with increased disability at 3 months. These results remained consistent in sensitivity analysis and across multiple endpoints. Our findings may have direct consequences for post-MT care of stroke patients.

Evidence on BP management after thrombectomy remains limited. Current guidelines provide recommendations based on low evidence and assign only weak strength of recommendation.^{6–8} Moreover, the guidelines primarily refer to BP during MT and observational data on postprocedural BP stability is scarce. The results of OPTIMAL-BP and ENCHANTED-2

indicate that active BP lowering in the first 24 hours can be harmful.^{13,14} These findings may help delineate lower BP limits that should be avoided. However, actively induced hypotension is not equivalent to spontaneously low BP, which may be tolerated well by many patients. Indeed, the ESO guideline contains an expert consensus statement, which suggests against induced hypertension.⁷ All in all, there might not be a generalizable lower BP threshold. Instead, an individual therapeutic range might be identified for every patient.

The INDIVIDUATE trial investigated this approach and compared the strategy of keeping systolic BP within ± 10 mm Hg from admission BP with a standard target range of 140–180 mm Hg during MT.²² However, the trial could not

Figure 2 Clinical Outcome at 3 Months

acOR = adjusted common odds ratio.

demonstrate any benefit of the intervention, which also held true for the DETERMINE trial (presented at ESOC 2025, results not published yet).²³ Still, the trial results do not rule out a potential advantage of an individualized approach.

Notably, both the INDIVIDUATE and DETERMINE trials focus(ed) on BP control during MT. However, this may not

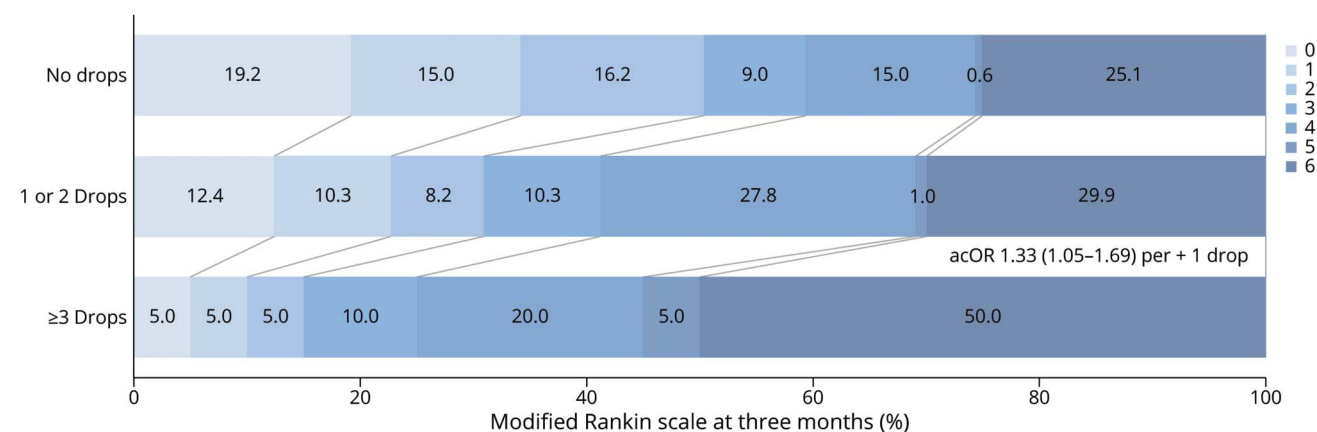
reflect the full impact of BP management. The potential harm caused by BP drops (infarct growth) or BP excess (ICH) may not be captured within the relatively short period of MT. Brain injury may be better understood as a cumulative process rather than the result of BP fluctuations limited to the time frame of MT. The results of ENCHANTED-2 and OPTIMAL-BP demonstrated that the first 24 hours after MT may

Table 3 Clinical and Safety Outcomes

	No BP drops (n = 177)	Any BP drops (n = 123)	a(c)OR (95 CI) Drops vs no drops	p Value	a(c)OR (95 CI) Per +1 drop	p Value
mRS at day 90, n (%)						
0	32 (19.2)	13 (11.1)	1.66 (1.02–2.70)	0.04	1.33 (1.05–1.69)	0.02
1	25 (15.0)	11 (9.4)				
2	27 (16.2)	9 (7.7)				
3	15 (9.0)	12 (10.3)				
4	25 (15.0)	31 (26.5)				
5	1 (0.6)	2 (1.7)				
6	42 (25.1)	39 (33.3)				
Missing	10	6				
mRS ≤2 at day 90, n (%)	84 (50.3)	33 (28.2)	0.35 (0.18–0.69)	<0.01	0.58 (0.40–0.84)	<0.01
mRS ≤3 at day 90, n (%)	99 (59.3)	45 (38.5)	0.41 (0.22–0.78)	<0.01	0.71 (0.51–0.98)	0.04
Early neurologic improvement, n (%)	49 (29.9)	31 (28.7)	0.99 (0.53–1.87)	0.98	0.96 (0.70–1.31)	0.78
Early neurologic deterioration, n (%)	22 (13.4)	24 (22.2)	2.22 (1.03–4.78)	0.04	1.73 (1.26–2.38)	<0.001
Any ICH, n (%)	36 (20.3)	31 (25.2)	1.50 (0.80–2.80)	0.20	1.31 (1.01–1.71)	0.04
sICH (ECASS-II), n (%)	11 (6.4)	6 (5.1)	0.94 (0.29–3.09)	0.92	1.37 (0.87–1.26)	0.18
In-hospital death, n (%)	27 (15.3)	26 (21.1)	1.66 (0.79–3.49)	0.18	1.31 (0.97–1.76)	0.08
Death at day 90, n (%)	42 (25.1)	39 (33.3)	1.25 (0.66–2.40)	0.50	1.24 (0.93–1.65)	0.14

Abbreviations: acOR = adjusted common odds ratio; BP = blood pressure; ECASS-II = European Cooperative Acute Stroke Study-II; ICH = intracranial hemorrhage; mRS = modified Rankin Scale; sICH = symptomatic ICH.

Figure 3 Clinical Outcome at 3 Months According to Number of Drops



acOR = adjusted common odds ratio.

be the most vulnerable phase. This finding is corroborated by our study, which provides robust evidence that BP drops may not only be harmful if actively induced in trials but also in real-life clinical practice, in which drops may occur both spontaneously and as a consequence of treatment.

We suggest that BP drops may worsen outcome primarily by reducing cerebral perfusion, particularly collateral flow. In line with this, we observed that approximately 20% of patients with BP drops had an ICA stenosis >70%, compared with 10% of those without drops. Such patients are highly dependent on collateral circulation and therefore at greatest risk of poor outcomes when BP declines occur. Automated BP monitoring algorithms aiming to detect precursors, such as increased BP variability, may help prevent drops through timely interventions, including positional adjustments or low-dose vasopressors in selected patients.

Regarding concurrent antihypertensive therapy, we noted a higher use of IV vasodilators in patients with BP drops, suggesting that at least some of the detected events might be treatment-related. As we found BP drops to be associated with worse outcome, our study supports the notion that initiation or escalation of IV antihypertensive medications should be undertaken very cautiously within the first 24 hours. The risks of reducing cerebral perfusion need to be balanced against the benefits of BP control, such as decreased rates of ICH. In our study, we found no overall association between BP drops and ICH rates. However, accounting for the number of drops, we found higher rates of any ICH as the drop burden increased. We hypothesize that recurrent BP drops may lead to repeated hypoperfusion, thereby accelerating infarct growth. As the latter has been linked to a higher risk of hemorrhagic transformation, this provides a plausible link between BP drops and the higher rates of any ICH we observed.²⁸ Notably, the association between BP drops and any ICH was no longer observed when analyses were restricted to patients with

mTICI 3. This finding supports the notion that particularly in patients with residual distal occlusions (mTICI 2b), BP drops may compromise collateral flow, thereby promoting infarct growth and subsequent hemorrhagic transformation.

When analyzing the subgroup of patients with mTICI 3 regarding clinical recovery, we noted that the detrimental effect of drops was even more pronounced than in the whole sample. This may point toward a relevant proportion of patients with impaired microcirculation despite successful reperfusion (no-reflow phenomenon). In these patients, sustained elevations in BP may represent a compensatory consequence. Persistently elevated BP may reflect a response to blood–brain barrier disruption, thromboinflammation, and early microvascular thrombosis, as discussed in Maier et al.²⁹ In these patients, BP drops may be particularly deleterious, as they might further compromise downstream microvascular perfusion and, as a consequence, lead to worse clinical outcomes.²⁹ By contrast, patients reaching both macrovascular and microvascular reperfusion may exhibit a spontaneous, clinically benign reduction in BP, consistent with restored autoregulatory function and associated with more favorable outcomes, as described in Carvalho Dias et al. and Anadani et al.^{30,31} Consequently, these findings do not contradict our results but point toward a need of individualized BP management, largely dependent on both macrovascular and microvascular reperfusion.

Although BP drops are explicitly addressed in current guidelines, they are not specifically defined.⁸ Accordingly, our study selected a threshold of 40 mm Hg as a cutoff, which inevitably represents an arbitrary choice. Sensitivity analyses demonstrated that even smaller BP drops (down to 20 mm Hg) may be associated with worse clinical outcomes if occurring repeatedly. In addition, BP variability, independent of discrete BP drop events, was a strong predictor of worse outcomes, a finding consistent with previous reports.^{32–34}

While our analysis focused on discrete BP drop events, overall BP stabilization (avoiding both rapid increases and decreases) should be targeted in future studies.

Previous observational studies investigated the impact of BP drops in patients undergoing MT and corroborated our findings.¹⁷⁻²⁰ However, while these studies focused mainly on intraprocedural BP variability, our study specifically quantified postprocedural BP drops within the first 24 hours.

Although the use of high-resolution BP data is a key strength of our study and the results are robust across multiple end points, limitations must be considered: The retrospective design makes our study prone to reporting bias. Reflecting real-world data, gaps of BP monitoring are present, and not all BP drops may have been captured. The parallel use of invasive and noninvasive BP monitoring may have introduced measurement bias, and the comparison of 30-minute averaged BP with noninvasive BP assessments remains an inherent methodological limitation. Limited use of invasive BP monitoring further restricted detection of abrupt BP fluctuations and precise estimation of BP drop rates in patients without arterial line. Using 30-minute averaged BP values may have further attenuated such fluctuations. Our study lacks data on collateral circulation, extent of perfusion mismatch and stroke volume, all potential modifiers of the interaction between BP drops and clinical outcome. Finally, associations are not causation. The causes of BP drops are various and may include both treatment-related factors (e.g., vasodilators, urinary catheterization for bladder retention) and systemic complications such as infection or cardiac failure. It remains to be clarified whether the BP drops themselves contribute to poor outcome or merely represent a marker of higher morbidity. Restricting future analyses to a more homogeneous patient population may help address this question.

In patients receiving MT for large vessel occlusion stroke, BP drops during the first 24 hours after groin puncture are common and associated with worse functional outcome. Preventing BP drops may represent a promising treatment target and the ongoing OPTIMAL-BP 2 trial (NCT06677970) will provide valuable insights.

Author Contributions

C. Riegler: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. A. Nelde: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. W.H. Bergström: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. R. Von Rennenberg: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. K. Bollweg: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Meisel: drafting/

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References

- Bangalore S, Schwamm L, Smith EE, et al. Blood pressure and in-hospital outcomes in patients presenting with ischaemic stroke. *Eur Heart J*. 2017;38(37):2827-2835. doi:10.1093/eurheartj/ehx330
- Aiyagari V, Gorelick PB. Management of blood pressure for acute and recurrent stroke. *Stroke*. 2009;40(6):2251-2256. doi:10.1161/STROKEAHA.108.531574
- Bath PM, Song L, Silva GS, et al. Blood pressure management for ischemic stroke in the first 24 hours. *Stroke*. 2022;53(4):1074-1084. doi:10.1161/STROKEAHA.121.036143
- Semplicini A, Maresca A, Boscolo G, et al. Hypertension in acute ischemic stroke: a compensatory mechanism or an additional damaging factor? *Arch Intern Med*. 2003;163(2):211-216. doi:10.1001/archinte.163.2.211
- Hilkens NA, Greving JP, Algra A, Klijn CJM. Blood pressure levels and the risk of intracerebral hemorrhage after ischemic stroke. *Neurology*. 2017;88(2):177-181. doi:10.1212/WNL.0000000000003489
- Powers WJ, Rabinstein AA, Ackerson T, et al. Guidelines for the early management of patients with acute ischemic stroke: 2019 update to the 2018 guidelines for the early management of acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2019;50(12):e344-e418. doi:10.1161/STR.0000000000000211
- Sandset EC, Anderson CS, Bath PM, et al. European Stroke Organisation (ESO) guidelines on blood pressure management in acute ischaemic stroke and intracerebral haemorrhage. *Eur Stroke J*. 2021;6(2):XLVIII-LXXXIX. doi:10.1177/23969873211026998
- Turc G, Bhogal P, Fischer U, et al. European Stroke Organisation (ESO) - European Society for Minimally Invasive Neurological Therapy (ESMINT) Guidelines on Mechanical Thrombectomy in Acute Ischaemic Stroke Endorsed by Stroke Alliance for Europe (SAFE). *Eur Stroke J*. 2019;4(1):6-12. doi:10.1177/2396987319832140

9. Malhotra K, Goyal N, Katsanos AH, et al. Association of blood pressure with outcomes in acute stroke thrombectomy. *Hypertension*. 2020;75(3):730-739. doi:10.1161/HYPERTENSIONAHA.119.14230
10. Goyal N, Tsivgoulis G, Iftikhar S, et al. Admission systolic blood pressure and outcomes in large vessel occlusion strokes treated with endovascular treatment. *J Neurointerv Surg*. 2017;9(5):451-454. doi:10.1136/neurintsurg-2016-012386
11. Mulder M, Ergezen S, Lingsma HF, et al. Baseline blood pressure effect on the benefit and safety of intra-arterial treatment in MR CLEAN (multicenter randomized clinical trial of endovascular treatment of acute ischemic stroke in the Netherlands). *Stroke*. 2017;48(7):1869-1876. doi:10.1161/STROKEAHA.116.016225
12. Maier B, Gory B, Taylor G, et al. Mortality and disability according to baseline blood pressure in acute ischemic stroke patients treated by thrombectomy: a collaborative pooled analysis. *J Am Heart Assoc*. 2017;6(10):e006484. doi:10.1161/JAHA.117.006484
13. Nam HS, Kim YD, Heo J, et al. Intensive vs conventional blood pressure lowering after endovascular thrombectomy in acute ischemic stroke: the OPTIMAL-BP randomized clinical trial. *JAMA*. 2023;330(9):832-842. doi:10.1001/jama.2023.14590
14. Yang P, Song L, Zhang Y, et al. Intensive blood pressure control after endovascular thrombectomy for acute ischaemic stroke (ENCHANTED2/MT): a multicentre, open-label, blinded-endpoint, randomised controlled trial. *Lancet*. 2022;400(10363):1585-1596. doi:10.1016/S0140-6736(22)01882-7
15. Mazighi M, Richard S, Lapergue B, et al. Safety and efficacy of intensive blood pressure lowering after successful endovascular therapy in acute ischaemic stroke (BP-TARGET): a multicentre, open-label, randomised controlled trial. *Lancet Neurol*. 2021;20(4):265-274. doi:10.1016/S1474-4422(20)30483-X
16. Mistry EA, Hart KW, Davis LT, et al. Blood pressure management after endovascular therapy for acute ischemic stroke: the BEST-II randomized clinical trial. *JAMA*. 2023;330(9):821-831. doi:10.1001/jama.2023.14330
17. Petersen NH, Ortega-Gutierrez S, Wang A, et al. Decreases in blood pressure during thrombectomy are associated with larger infarct volumes and worse functional outcome. *Stroke*. 2019;50(7):1797-1804. doi:10.1161/STROKEAHA.118.024286
18. Valent A, Sajadhoussen A, Maier B, et al. A 10% blood pressure drop from baseline during mechanical thrombectomy for stroke is strongly associated with worse neurological outcomes. *J Neurointerv Surg*. 2020;12(4):363-369. doi:10.1136/neurintsurg-2019-015247
19. Löwhagen Hendén P, Rentzos A, Karlsson JE, et al. Hypotension during endovascular treatment of ischemic stroke is a risk factor for poor neurological outcome. *Stroke*. 2015;46(9):2678-2680. doi:10.1161/STROKEAHA.115.009808
20. Whalin MK, Halenda KM, Haussen DC, et al. Even small decreases in blood pressure during conscious sedation affect clinical outcome after stroke thrombectomy: an analysis of hemodynamic thresholds. *AJNR Am J Neuroradiol*. 2017;38(2):294-298. doi:10.3174/ajnr.A4992
21. Maier B, Fahed R, Khoury N, et al. Association of blood pressure during thrombectomy for acute ischemic stroke with functional outcome: a systematic review. *Stroke*. 2019;50(10):2805-2812. doi:10.1161/STROKEAHA.119.024915
22. Chen M, Meis J, Potreck A, et al. Effect of individualized versus standardized blood pressure management during endovascular stroke treatment on clinical outcome: a randomized clinical trial. *Stroke*. 2023;54(11):2755-2765. doi:10.1161/STROKEAHA.123.044062
23. Maier B, Gory B, Chabanne R, et al. Effect of an individualized versus standard blood pressure management during mechanical thrombectomy for anterior ischemic stroke: the DETERMINE randomized controlled trial. *Trials*. 2022;23(1):S98. doi:10.1186/s13063-022-06538-9
24. Alegiani AC, Dorn F, Herzberg M, et al. Systematic evaluation of stroke thrombectomy in clinical practice: the German Stroke Registry Endovascular Treatment. *Int J Stroke*. 2019;14(4):372-380. doi:10.1177/1747493018806199
25. Riegler C, von Rennenberg R, Bollweg K, et al. Endovascular therapy in patients with internal carotid artery occlusion and patent circle of Willis. *J Neurointerv Surg*. 2024;16(7):644-651. doi:10.1136/jnis-2023-020556
26. Wollenweber FA, Tiedt S, Alegiani A, et al. Functional outcome following stroke thrombectomy in clinical practice. *Stroke*. 2019;50(9):2500-2506. doi:10.1161/STROKEAHA.119.026005
27. Nelde A, Klammer MG, Nolte CH, et al. Data lake-driven analytics identify nocturnal non-dipping of heart rate as predictor of unfavorable stroke outcome at discharge. *J Neurol*. 2023;270(8):3810-3820. doi:10.1007/s00415-023-11718-x
28. Méot M, Munsch F, Lapergue B, et al. Hemorrhagic transformation after endovascular treatment: baseline infarct volume is a better predictor than infarct growth rate. *Eur Stroke J*. Published online July 15, 2025. doi:10.1177/23969873251357151
29. Maier B, Brauner R, Escalard S, et al. Association of contrast enhancement after reperfusion with outcomes according to blood pressure lowering in patients with acute ischemic stroke. *Neurology*. 2022;99(21):e2385-e2394. doi:10.1212/WNL.000000000000201173
30. Carvalho Dias M, Gabriel D, Saraiva M, et al. Spontaneous systolic blood pressure drop early after mechanical thrombectomy predicts dramatic neurological recovery in ischaemic stroke patients. *Eur Stroke J*. 2020;5(4):362-369. doi:10.1177/2396987320933384
31. Anadani M, Arthur AS, Alawieh A, et al. Blood pressure reduction and outcome after endovascular therapy with successful reperfusion: a multicenter study. *J Neurointerv Surg*. 2020;12(10):932-936. doi:10.1136/neurintsurg-2019-015561
32. Maier B, Gory B, Lapergue B, et al. Effect of blood pressure variability in the randomized controlled BP TARGET trial. *Eur J Neurol*. 2022;29(3):771-781. doi:10.1111/ene.15194
33. Kim TJ, Park HK, Kim JM, et al. Blood pressure variability and hemorrhagic transformation in patients with successful recanalization after endovascular recanalization therapy: a retrospective observational study. *Ann Neurol*. 2019;85(4):574-581. doi:10.1002/ana.25434
34. Palaiodimos L, Joundi RA, Katsanos AH, et al. Association between blood pressure variability and outcomes after endovascular thrombectomy for acute ischemic stroke: an individual patient data meta-analysis. *Eur Stroke J*. 2024;9(1):88-96. doi:10.1177/23969873231211157